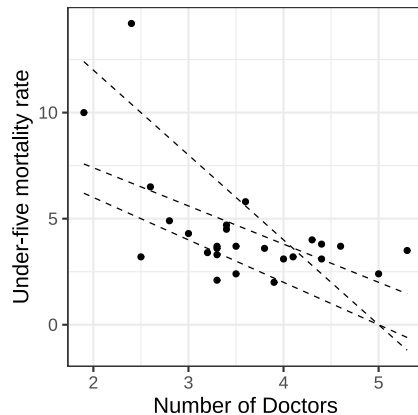


Simple Linear Regression

Simple Linear Regression

- This is a very brief introduction to simple linear regression.
- A full, in-depth treatment is provided in EDUC 642.
- Simple linear regression and correlation are closely connected concepts.
- The goal of the simple linear regression is to identify a line that best summarizes the relationship between two variables.



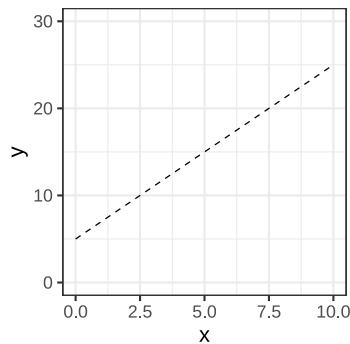
- There are infinitely many lines we could draw.
- Which line best represents the relationship?
- Regression finds the line that minimizes prediction error (least squares).

What is a line?

- A line represents a mathematical function that maps an input variable x to an output variable y .

$$f(x) = y = b + mx$$

- There are two key coefficients in this equation:
 - m : slope of the line
 - b : intercept
- The intercept (b) represents the point where the line crosses on the y-axis.
- The slope (m) represents the amount of change in y for every one unit increase in x .

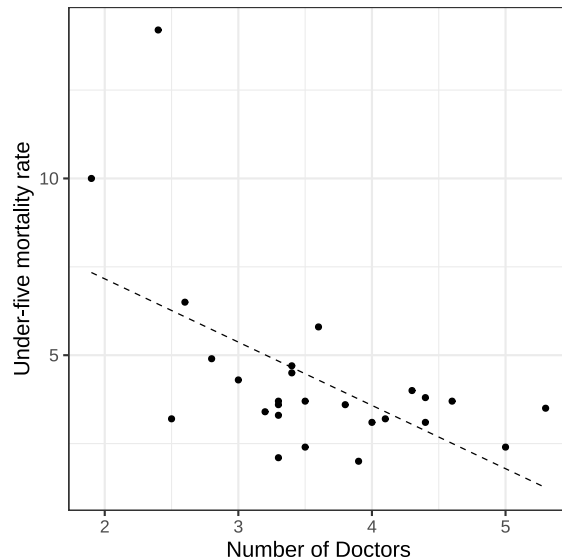


$$y = 5 + 2x$$

x	y
0	5
1	7
2	9
3	11

The goal of simple linear regression is to find the line that best fits the observed data, modeling the relationship between:

- a predictor variable (X)
- an outcome variable (Y)



$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

- $\hat{\beta}_0$: estimated intercept
- $\hat{\beta}_1$: estimated slope
- X_i : observed value of variable X for the i^{th} observation
- $\hat{Y}_i$: predicted value of variable Y for the i^{th} observation

We will not cover the derivation of $\hat{\beta}_0$ and $\hat{\beta}_1$ here. Those details are covered in depth in EDUC 642.

R Implementation

```
# Use the lm() function
d <- read.csv('B:/UO Teaching/EDUC614/Winter22/Week9/infant_mortality.csv', header=TRUE)

lm(formula = infmor ~ 1 + doctor,
   data     = d)
```

```
Call:
lm(formula = infmor ~ 1 + doctor, data = d)
```

```
Coefficients:
(Intercept)      doctor
    10.735         -1.793
```

- The `lm()` function returns two values. The first one is the intercept estimate ($\hat{\beta}_0 = 10.735$) and the second one is the slope estimate ($\hat{\beta}_1 = -1.793$).
- How do you interpret these numbers?
- What is the model predicted number of under-five mortality rate (per 1,000 live births) for a country with 4 doctor per 1,000 inhabitants?

Statistical Inference for Regression Coefficients

- The estimated coefficients, $\hat{\beta}_0$ and $\hat{\beta}_1$, are based on sample data, not the population.
- Due to sampling variability, they may differ from zero just by chance.
- We test whether these coefficients are statistically different from zero in the population (i.e., reflect a real relationship, not random noise).

The null hypothesis for the two coefficients are

$$H_0 : \beta_0 = 0 \text{ vs. } H_A : \beta_0 \neq 0$$

$$H_0 : \beta_1 = 0 \text{ vs. } H_A : \beta_1 \neq 0$$

- We test these two hypotheses with a t-test (df = N-2).
- If we reject H_0 , it indicates that the coefficient is statistically significant.
- These results can be obtained using the `lm()` function in R.

```
# Save the output as an R object in your environment
my.model <- lm(formula = infmor ~ 1 + doctor,
               data    = d)

# Summarize the output
summary(my.model)
```

```
Call:
lm(formula = infmor ~ 1 + doctor, data = d)

Residuals:
    Min       1Q   Median       3Q      Max
-3.0521 -1.1924 -0.2518  0.8742  7.7686

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)   10.735      1.967    5.458 1.31e-05 ***
doctor        -1.793      0.538   -3.333  0.00278 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.172 on 24 degrees of freedom
Multiple R-squared:  0.3164,    Adjusted R-squared:  0.2879
F-statistic: 11.11 on 1 and 24 DF,  p-value: 0.002779
```

Effect Size for Simple Linear Regression

- In R, the `lm()` model output reports **Multiple R-squared**
- R^2 quantifies effect size. It reflects the strength of the relationship between the predictor (X) and outcome (Y).
 - For a simple linear regression with one predictor, it is equivalent of correlation between two variables.
- It can be interpreted as “the proportion of variance in the Y that can be explained/predicted by X .”
 - $R^2 = 0.62$, 62% of the variance in Y is explained by X .

Reporting the results of Simple Linear Regression

```
summary(my.model)
```

```
Call:
lm(formula = infmor ~ 1 + doctor, data = d)

Residuals:
    Min       1Q   Median       3Q      Max
-3.0521 -1.1924 -0.2518  0.8742  7.7686

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   10.735      1.967   5.458 1.31e-05 **
doctor        -1.793      0.538  -3.333  0.00278 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.172 on 24 degrees of freedom
Multiple R-squared:  0.3164,    Adjusted R-squared:  0.2879
F-statistic: 11.11 on 1 and 24 DF,  p-value: 0.002779
```

	Coefficient	Std.Error	t	p
Intercept	10.74	1.97	5.46	<.001
Slope (doctor)	-1.79	0.54	-3.33	0.003

The unstandardized regression coefficient for the number of doctors per 1,000 inhabitant was -1.79 and statistically significant, $t(24) = -3.33$, $p = 0.003$, indicating that for each additional doctor for 1,000 inhabitants, the number of under-five mortality rate (per 1,000 live births) is reduced by 1.79 in a country. The number of doctors per 1,000 inhabitants explains about 31.6% of the variability in under-five mortality rate across 26 countries.